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# Chapter 7
# Hypothesis test

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- Basic concepts and methods
- Hypothesis Testing of Normal Population Mean
- Hypothesis Testing of Normal Population variance

# Introduction

In practical work, we often encounter such problems:

(1) Has the therapeutic effect of a certain drug improved after improving the process?

(2) Is it true to assume that the population follows a certain distribution?

How to make a judgment on the above issues through sampling samples?

# Introduction

At this point, the following steps are often taken:

1. make an appropriate assumption,
2. conduct experiments to obtain statistical samples,
3. construct appropriate statistical methods,
4. decide whether to accept the original assumption.

Hypothesis testing is a statistical inference method that makes a judgment on whether a hypothesis is accepted or rejected based on the information provided by the sample

# Section 1 Basic concepts and methods for Hypothesis test

# Basis

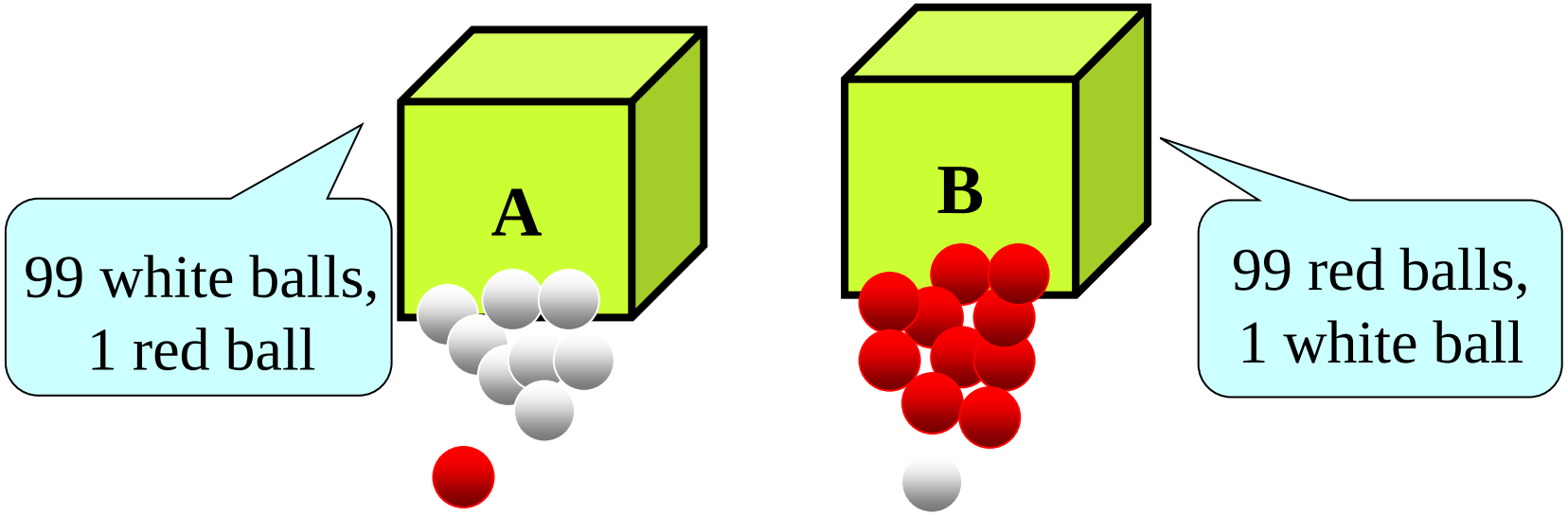
## Basis for judging "hypothesis"

**Small probability events are almost impossible to occur in one test.**

If an event is a small probability event under the condition that hypothesis $H_0$ holds, but it occurs in a test, then the correctness of hypothesis $H_0$ can be doubted and $H_0$ can be rejected.

# Basis

**eg:** There are two boxes, each containing 100 balls.




Now take out one ball randomly from the two boxes.

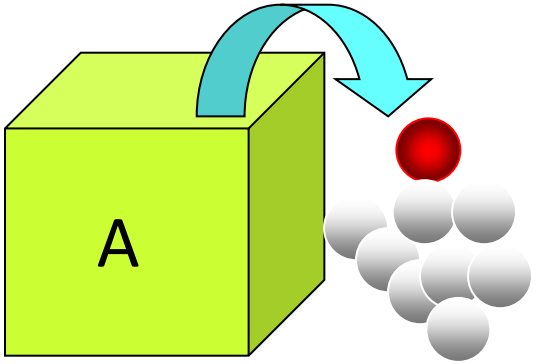
Q: Is this ball from box A or box B?

# Basis

**Suppose:** this ball is taken from box A.

**Sampling:** When a random ball is found to be a red ball,

Is this hypothesis true?




Suppose there are 99 white balls, and the probability of take out the red ball is only 1 / 100.

**Rejection:** However, this small probability event actually occurred in an experiment, so we have to doubt this hypothesis.

# Basis

If a small probability event actually occurs in an experiment, you can reject the original hypothesis with great certainty, otherwise you can't reject the original hypothesis.

In hypothesis testing, this small probability is often called the significance level, expressed by $\alpha$.

# 7.1.1 Basic concepts

Hypothesis testing is the process of making possible hypotheses about the type of distribution function or uncertain parameters in a population, and conducting reasonable methods to determine whether these hypotheses are acceptable based on the samples.

In statistics,

- a part of a statistical hypothesis is called a ***null hypothesis*** (or ***original hypothesis***), denoted as $H_0$;
- The opposite of the hypothesis is called the ***alternative hypothesis***, denoted as $H_1$.

# 7.1.1 Basic concepts

The area where the sample observations that were rejected under the null hypothesis $H_0$ are located is called the ***rejection domain,*** which is a subset of the sample space, denoted as $W$, and its complement $\overline{W}$ is called the ***acceptance domain***.

**Principle of small probability**

Small probability events are almost impossible to occur in an experiment /one test.

# 7.1.1 Basic concepts

## The basic idea of hypothesis testing:

Contrary proof of probabilistic properties:

- To test hypothesis $H_0$, first, assume that $H_0$ is correct,
- construct a small probability event $\boldsymbol{A}$, then test whether the small probability event $\boldsymbol{A}$ occurred in an experiment based on the conditions given by the problem,
- If event $\boldsymbol{A}$ actually occurs, it indicates that the sampling result does not match hypothesis $H_0$,
- According to the principle of “small probability events almost do not occur in one test", reject hypothesis $H_0$,
- Otherwise, accept hypothesis $H_0$.

# 7.1.2 Basic methods

Example 1: A certain factory has a batch of 10000 products that need to be inspected and qualified before its shipped. According to regulations, the defect rate should not exceed 2%, now we taken 100 samples from it and 5 defective products were found. Can this batch of products be shipped? （$\alpha = 0.05$)

Analysis:

Based on intuition, this batch of products cannot be shipped, cause the defect rate obtained from sampling is 5%.

However, due to the randomness of the sample, how can we determine whether the overall (all products) defect rate is<2% based on the sampling results??

# 7.1.2 Basic methods

Example 1: A certain factory has a batch of 10000 products that need to be inspected and qualified before its shipped. According to regulations, the defect rate should not exceed 2%, now we taken 100 samples from it and 5 defective products were found. Can this batch of products be shipped? （α= 0.05)

Solution:

1. Propose a Hypothesis

$H_0$: The overall defect rate $p \leq 0.02$; $H_1$: $p > 0.02$

2. Conduct an experiment

let $X_i = \begin{cases} 1, & \text{the } i^{\text{th}} \text{ sample is a defective product} \\ 0, & \text{otherwise} \end{cases}$

that is, $X_i \sim B(1, p)$ (i=1,2,3,..., 100),

Let $X = \sum X_i$ , that is, $X$ assumed to be the number of defectives in the 100 sampled products.

Thus, $X \sim B(100, p)$

# 7.1.2 Basic methods

Example 1: A certain factory has a batch of 10000 products that need to be inspected and qualified before its shipped. According to regulations, the defect rate should not exceed 2%, now we taken 100 samples from it and 5 defective products were found. Can this batch of products be shipped? （α= 0.05)

Solution:

3. Assuming $H_0$ ($p \leq 0.02$) holds, then:

$$P(X=5;p)= C_{100}^{5} p^{5}(1-p)^{95} = f(p)$$

$$f(p)' = C_{100}^{5} p^{4}(1-p)^{94}(5-100p)$$

*So, when* $p \leq 2\%$, $f(p)\uparrow$

$$f(p) \leq f(0.02) = 0.035 < \alpha = 0.05$$

# 7.1.2 Basic methods

Example 1: A certain factory has a batch of 10000 products that need to be inspected and qualified before its shipped. According to regulations, the defect rate should not exceed 2%, now we taken 100 samples from it and 5 defective products were found. Can this batch of products be shipped? （α= 0.05)

Solution:

4. Make an inference

- Under the Assuming that $H_0$ ($p \leq 0.02$) holds, $\{X=5\}$ is a small probability event;
- but this small probability event actually occurred in ONE test, which violates the principle of small probability
- it is unreasonable, thus the null hypothesis $H_0$ should be denied, that is, $p > 0.02$,
- So, this batch of products can't be shipped.

# 7.1.2 假设检验的方法

## 假设检验的两类错误：

由于样本具有随机性，小概率事件也可能发生，所以假设检验所得出的决策并不总是正确的.

| 客观事实 | 决策结果 | | |
|---|---|---|---|
| 原假设$H_0$为真 | 拒绝$H_0$ → | 错误 | 这种错误称为第一类错误<br>又称“弃真”错误.<br>犯第一类错误的概率记为$\alpha$. |
| | 接受$H_0$ → | 正确 | |
| 原假设$H_0$为假 | 接受$H_0$ → | 错误 | 这种错误称为第二类错误<br>又称“取伪”错误.<br>犯第二类错误的概率记为$\beta$. |
| | 拒绝$H_0$ → | 正确 | |

# 7.1.2 假设检验的方法

| 客观事实 | 决策结果 | | |
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| | 拒绝$H_0$ → | 正确 | |

注 1°当样本容量 $n$ 一定时, 若减少犯第一类错误的概率, 则犯第II类错误的概率往往增大. 2°若要使犯两类错误的概率都减小, 除非增加样本容量.

# 7.1.2 假设检验的方法

假设检验的基本步骤：

（1）建立假设：常把一个被检验的假设作为原假设，用$H_0$表示。

（2）选择检验统计量，给出原假设的拒绝域形式

由样本对原假设进行判断总是通过一个统计量完成，该统计量称为检验统计量。使原假设$H_0$被拒绝的样本观测值所在区域构成了拒绝域$W$.

（3）作出判断

由样本的观测值计算检验统计量的观测值，进而观察“小概率事件”是否发生。若样本观测值落入拒绝域内，则拒绝原假设$H_0$，否则接受$H_0$。